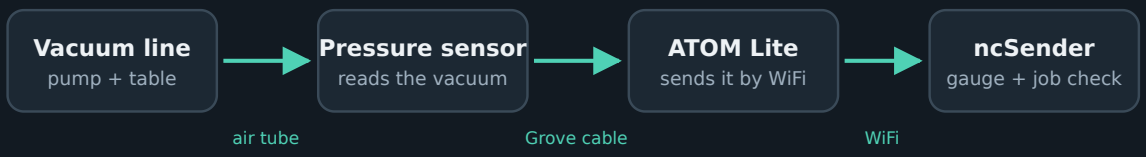
INSTALL GUIDE • FOR A CNC VACUUM TABLE

Vacuum Gauge for ncSender

For a CNC vacuum table. A small WiFi sensor on the vacuum line shows how hard the table is holding, live inside ncSender, and can warn you or pause the job if the vacuum drops.



IN THIS PACKAGE

PDF

Vacuum Gauge - Install Guide.pdf

This guide.

INO

VacGauge / VacGauge.ino

Program for the ATOM Lite. Open with Arduino IDE.

ZIP

vacuum-gauge-ncsender-plugin.zip

Plugin for ncSender. Install it as it is. Do not unzip.

PARTS YOU NEED

#	PART	QTY	NOTES
1	**M5Stack ATOM Lite** (ESP32)	1	The small WiFi computer. About £8.
2	**M5Stack Tube / Air Pressure Unit** (U130)	1	The sensor. Grove cable included. About £7.
3	USB-C cable + 5 V USB power adapter	1	A data cable, not charge-only.
4	Silicone tube, 2.5 mm inside × 4 mm outside	about 5 cm	Stretches over the 3 mm sensor nipple.
5	Push-in reducer, 1/4" to 5/32" (4 mm)	1	Joins the small tube to the 1/4" tube.
6	1/4" outside diameter PU air tube	as needed	Vacuum line to sensor. Keep it short.
7	Push-in tee, 1/4"	1	Goes into the vacuum line.
8	M5Stack Atomic Battery Base (200 mAh)	optional	Backup power only.

ALSO NEEDED

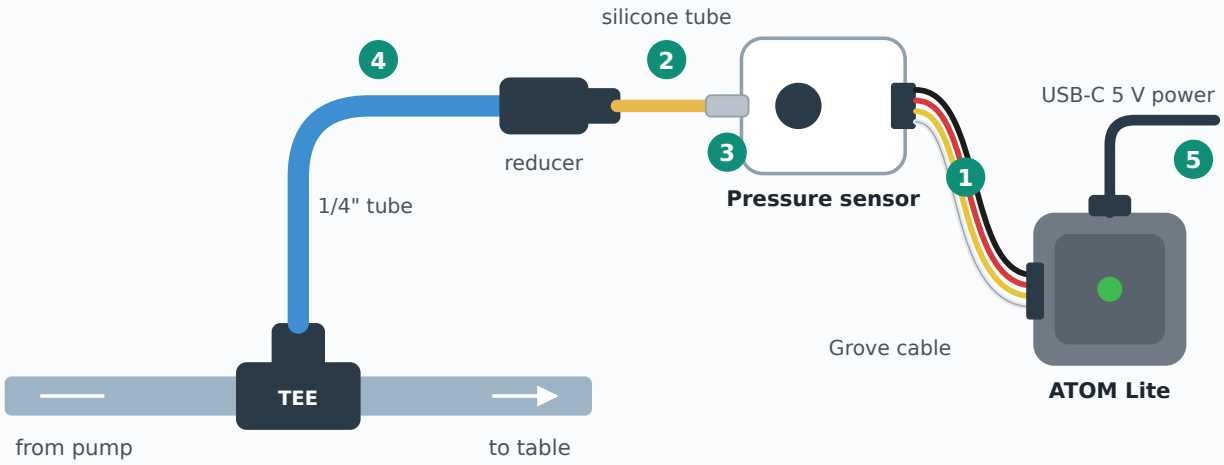
A computer with **Arduino IDE** (free, arduino.cc) to program the ATOM once • shop WiFi name and password (2.4 GHz) • **ncSender 2.0.225 or newer** on the CNC computer, on the same network.

> **Do not use a 0–40 kPa sensor** (the small red HX710B boards). It only reaches about 12 inHg. A vacuum table runs at 15 to 25 inHg.

PART 1

Connect the hardware

No soldering. Everything pushes together.



The numbers match the steps below.

1. **Grove cable:** plug one end into the sensor and the other into the Grove port on the ATOM Lite. The plug fits only one way.
2. **Silicone tube into the reducer:** push one end of the small silicone tube into the 5/32" (small) side of the reducer.
3. **Silicone tube onto the sensor:** stretch the other end over the metal nipple on the sensor. Push it on fully.
4. **1/4" tube:** push the 1/4" tube into the big side of the reducer and run it to the tee in your vacuum line.
5. **Power:** plug the ATOM Lite into USB power, or take 5 V from the SLB-EXT (next page). For programming in Part 2, plug it into the computer instead.

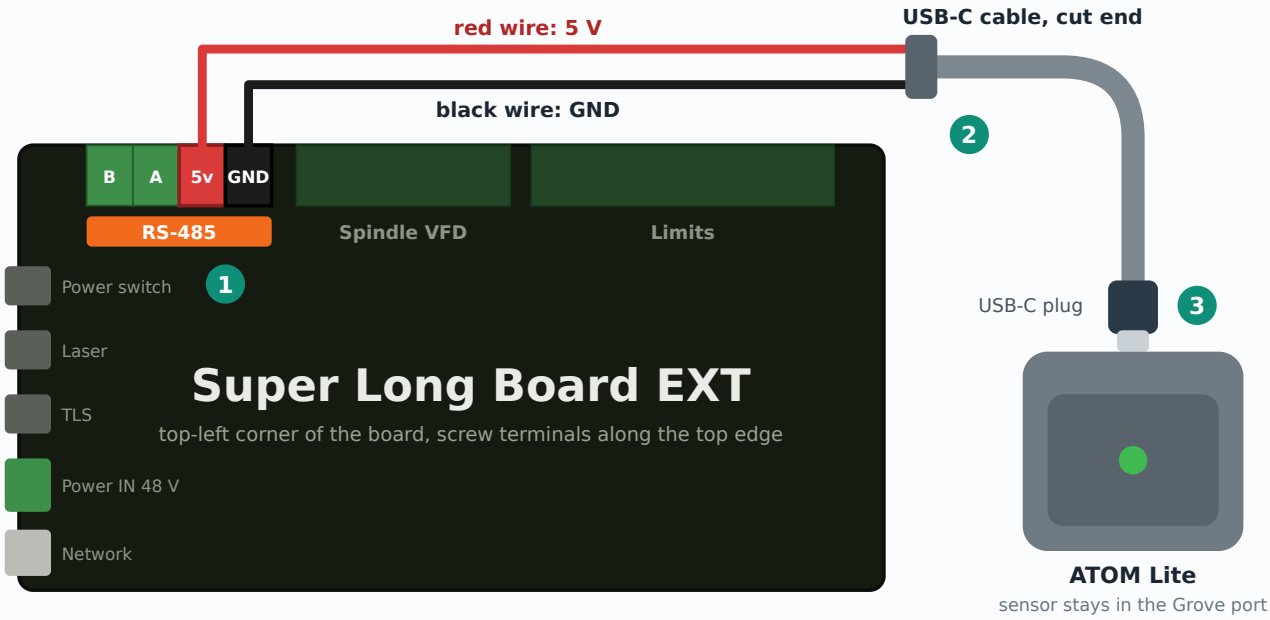
Where to put the tee. Between the pump and the table valve to read the pump. One sensor per table, after each valve, to read each table.

Keep the tube short and straight. A pinched or very long tube gives a slow or wrong reading. If the silicone leaks in the reducer, use stiffer 2.5 × 4 mm PVC tube.

POWER

Power the ATOM Lite

The ATOM Lite needs 5 V through its USB-C port. Use a USB power adapter, or take 5 V from the SLB-EXT so the gauge switches on with the machine.



5 V and GND from the RS-485 terminal on the SLB-EXT, through a USB-C cable, into the ATOM Lite.

FROM THE SLB-EXT

1. **Find the RS-485 terminal** at the top-left of the SLB-EXT. It has four screws marked **B**, **A**, **5v**, **GND**. B and A are for the spindle VFD. Leave them as they are.
2. **Prepare a USB-C cable.** Use a USB-C power cable with two bare wires, or cut the other plug off a spare USB-C cable and strip the **red** and **black** wires. Tape off any other wires. With the machine off, screw **red into 5v** and **black into GND**.
3. **Plug the USB-C end into the ATOM Lite** and switch the machine on. The ATOM's light comes on.

Check before you plug in. Wire colours in a cut cable can differ. With a multimeter, confirm 5 V between the two wires and which one is plus, before the plug goes into the ATOM. Never use a 24 V or 48 V terminal.

If the ATOM restarts or drops WiFi, the 5 V pin is not giving enough current. Use a USB power adapter instead. To program the ATOM, unplug this cable and use a normal USB-C cable to the computer.

PART 2

Program the ATOM Lite

You do this once, with Arduino IDE on any computer.

1. **Install Arduino IDE** from arduino.cc and open it.
2. **Add ESP32 boards:** open **Tools** → **Board** → **Boards Manager**, search `esp32`, install **esp32 by Espressif Systems**.
   Not in the list? Open **File** → **Preferences** and paste this into "Additional boards manager URLs", then try again:
   `https://espressif.github.io/arduino-esp32/package_esp32_index.json`
3. **Open the program:** **File** → **Open** and choose `VacGauge.ino`. Keep it inside the folder named `VacGauge`.
4. **Type your WiFi name and password** near the top of the file, between the quote marks:

```
// ---------------- settings you change ----------------
const char* WIFI_SSID     = "YOUR_WIFI_NAME";
const char* WIFI_PASSWORD = "YOUR_WIFI_PASSWORD";
const char* HOSTNAME      = "vac-gauge";      // -> http://vac-gauge.local
```

   Capital and small letters matter. The WiFi must be 2.4 GHz.
5. **Plug the ATOM Lite into the computer** with the USB-C cable.
6. **Choose the board and port:** **Tools** → **Board** → **esp32** → **M5Atom**, then **Tools** → **Port** → the new port that appeared.
7. **Click Upload** (the arrow button). Wait for "Done uploading". The first time takes a few minutes.
8. **Test it:** on any phone or computer on the same WiFi, open `http://vac-gauge.local`. You see the reading. With the pump off it is close to 0.

WHAT THE LIGHT ON THE ATOM MEANS

Green	Yellow	Red	Blinking blue	Blinking purple
Vacuum is good (15 inHg or more)	Marginal (8 to 15 inHg)	Too weak, or pump off	No WiFi. Check name and password	Sensor not plugged in

If something goes wrong

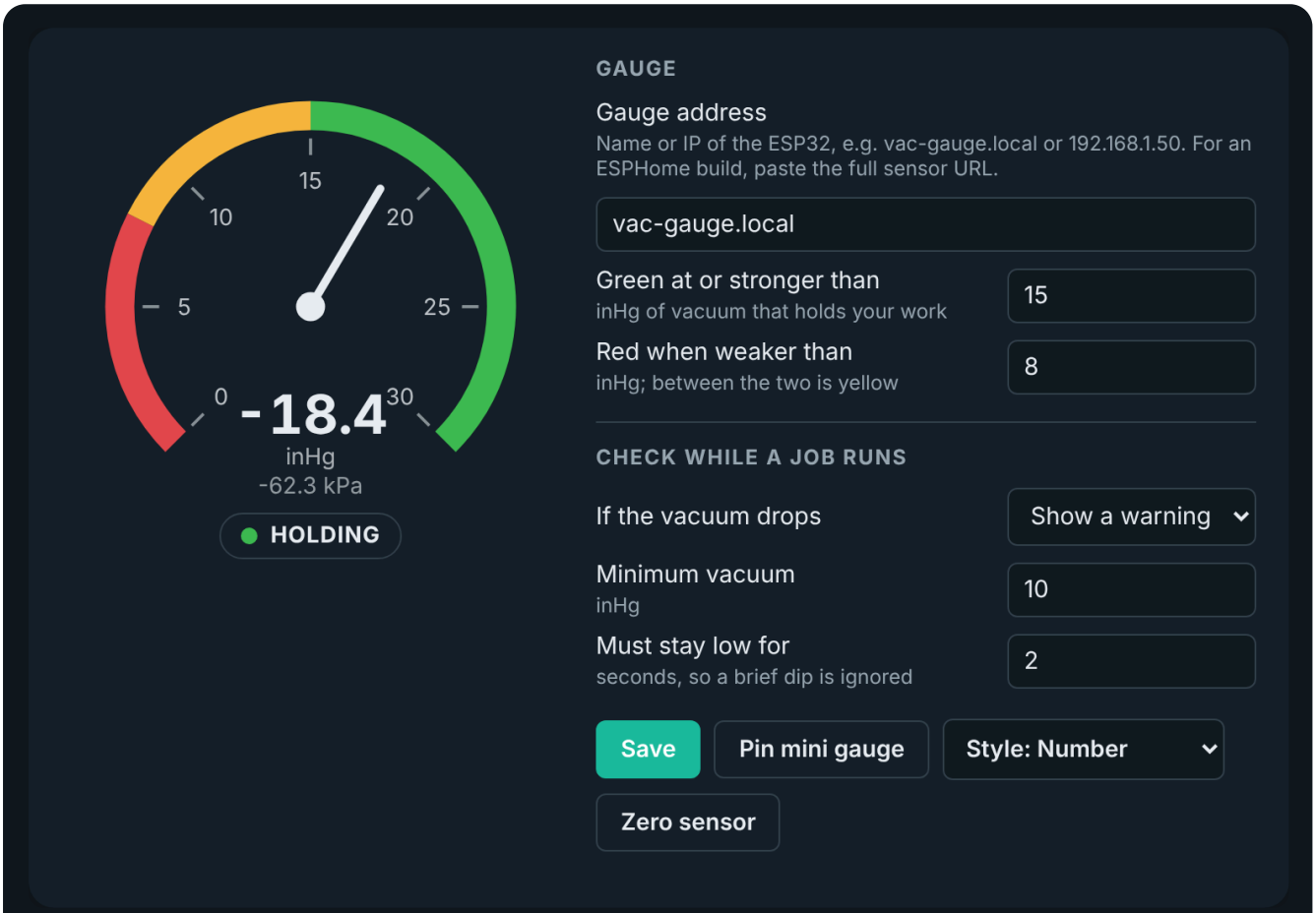
No port appears: try another USB-C cable. Many cables only charge.
Upload fails: set **Tools** → **Upload Speed** → **115200** and try again.
`vac-gauge.local` does not open: open **Tools** → **Serial Monitor** at 115200 and press the small reset button on the side of the ATOM. It prints the IP address, for example `192.168.1.50`. Use that number instead.

PART 3

Install the plugin in ncSender

Use the file `vacuum-gauge-ncsender-plugin.zip` as it is. Do not unzip it.

1. **Save the zip** to a normal folder on the CNC computer, such as Desktop or Downloads.
2. In ncSender open the **Plugins** window and click **Install from a ZIP file**.
3. Click **Choose File**, pick the zip, then click **Install from File**.
4. **Vacuum Gauge** now shows in the plugin list, switched on.
5. Open **Tools** → **Vacuum Gauge**. You see this window:



The Vacuum Gauge window: live dial on the left, settings on the right.

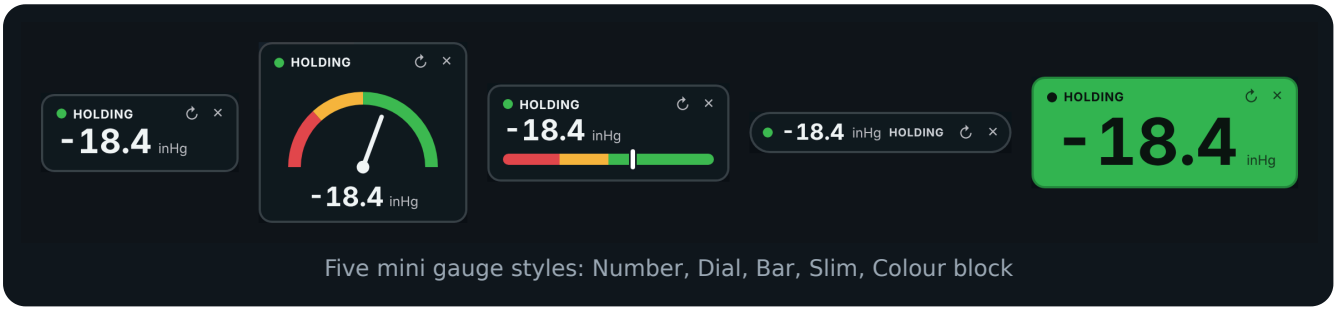
6. **Gauge address:** leave `vac-gauge.local`. If the dial says NO SIGNAL, type the ATOM's IP address instead (for example `192.168.1.50`) and click **Save**.

> **"Failed to fetch" when installing?** Save the zip to Desktop first, then choose it again. If it still fails, install by hand: unzip it into a folder named `community.vacuum-gauge`, put that folder into ncSender's `plugins` folder, restart ncSender, and switch the plugin on.
>
> **Windows** `%APPDATA%\ncSender\plugins\`
> **Mac** `~/Library/Application Support/ncSender/plugins/`
> **Linux / Pi** `~/.config/ncSender/plugins/`

PART 4

Set it up and use it

1. **Zero the sensor.** Pump off, tube open to air. In the Vacuum Gauge window click **Zero sensor**. The dial now reads 0.
2. **Find your numbers.** Turn the pump on and hold down a real workpiece. Note the reading. Compare it with a mechanical gauge if you have one.
3. **Set the colours.** "Green at or stronger than" = the vacuum that holds your work well. "Red when weaker than" = the vacuum where parts can move. Click **Save**.
4. **Pin the mini gauge.** Click **Pin mini gauge**. A small reading stays on the main screen. Drag it where you want it. Choose its look in the **Style** list. After restarting ncSender, open this window once to bring it back.



Five mini gauge styles: Number, Dial, Bar, Slim, Colour block

HOLDING (green): good vacuum. **MARGINAL** (yellow): between your two levels. **TOO WEAK** (red): weaker than the red level. **NO SIGNAL** (grey): ncSender cannot reach the gauge.

CHECK WHILE A JOB RUNS

Choose one, then click Save. "Do nothing" turns it off.

SETTING	WHAT IT DOES
Show a warning (default)	A message appears if the vacuum drops during a job. The job keeps running.
Feed hold + warning	The machine stops moving and a message appears. Fix the vacuum, then press Resume.
Minimum vacuum and **Must stay low for**	The check acts when the vacuum is weaker than the minimum (default 10 inHg) for this many seconds (default 2), so a short dip is ignored.

> **Safety.** Feed hold stops the movement but the **spindle keeps turning**. This gauge helps the operator. It is not a safety device and it does not replace watching the machine. If the gauge loses WiFi during a job you get a warning, and the job is not paused.

PROBLEMS

WHAT YOU SEE	WHAT TO DO
Dial says NO SIGNAL	Check the ATOM has power and its light is not blinking blue. Type the ATOM's IP address in "Gauge address" and Save.
Not 0 with pump off	Click Zero sensor with the pump off and the tube open to air.
Slow or low reading	Look for a pinched tube or a leak at the silicone tube.